WORKED SOLUTION

STEP I 1987, Question 11

Mechanics • Circular Motion and Energy

STEP I 1987, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q11

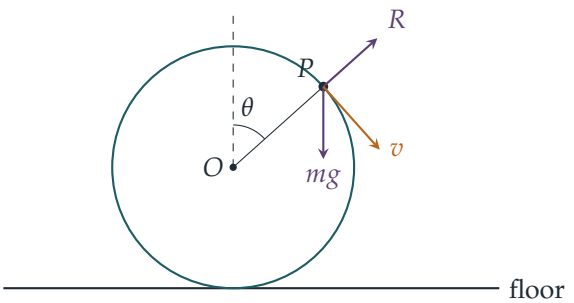
The key idea

While the particle is on the sphere it moves in a circle, so two equations govern it: energy conservation gives its speed, and Newton's second law along the radius gives the normal reaction. Contact is lost exactly when the reaction falls to zero. After that the particle is a projectile, so its horizontal velocity never changes; combining that with energy conservation all the way down to the floor gives the vertical velocity without solving for the trajectory.

Solution

Model

Treat the particle as a point mass on a smooth, fixed sphere with centre O, a height r above the floor (the sphere rests on the floor). "Displaced gently" means it starts at the top with negligible speed. Let θ be the angle between OP and the upward vertical, where P is the particle, and let v be its speed. The only forces are the weight mg and the normal reaction R along OP, outwards.



Where contact is lost

The particle has fallen a height $r - r\cos\theta$, so by conservation of energy (the reaction does no work, being perpendicular to the motion)

$$\tfrac{1}{2}mv^2 = mgr(1-\cos\theta), \qquad \text{so} \qquad v^2 = 2gr(1-\cos\theta).$$

While in contact the particle moves on a circle of radius r, so its acceleration towards O is v^2/r. Resolving towards O:

$$mg\cos\theta - R = \frac{mv^2}{r} = 2mg(1-\cos\theta), \qquad \text{so} \qquad R = mg(3\cos\theta - 2).$$

At the top $R = mg > 0$, and R decreases as θ increases. Contact is lost when $R = 0$, that is when

$$\cos\theta = \tfrac{2}{3}, \qquad v^2 = \tfrac{2}{3}gr.$$

Ask yourself Which angle is the angle the velocity makes with the horizontal?

The velocity is along the tangent to the circle, which is perpendicular to the radius OP. The radius makes angle θ with the vertical, so the tangent makes the same angle θ with the horizontal (turn both lines through a right angle). So at the moment it leaves the sphere the particle is moving downwards at an angle

$$\cos^{-1}\tfrac{2}{3} \approx 48.2^\circ$$

below the horizontal.

Striking the floor

At the moment of leaving, the horizontal component of velocity is

$$v\cos\theta = \sqrt{\tfrac{2}{3}gr}\cdot\tfrac{2}{3}, \qquad \text{so} \qquad (v\cos\theta)^2 = \tfrac{8}{27}gr.$$

From then on the only force is gravity, so this horizontal component is unchanged until the particle reaches the floor. (The particle does not touch the sphere again: once R would be negative the sphere is curving away faster than the particle falls, and the particle lands at a horizontal distance of about $1.46r$ from the line through O, well clear of the sphere.)

Now apply energy conservation to the whole motion, from rest at the top (height $2r$) down to the floor. The particle hits the floor with speed V where

$$\tfrac{1}{2}mV^2 = mg(2r), \qquad \text{so} \qquad V^2 = 4gr.$$

If V_y is the vertical component of velocity on impact, then $V^2 = (v\cos\theta)^2 + V_y^2$, so

$$V_y^2 = 4gr - \tfrac{8}{27}gr = \tfrac{100}{27}gr, \qquad V_y = \frac{10}{3}\sqrt{\frac{gr}{3}}.$$

The vertical component of momentum as it strikes the floor therefore has magnitude

$$mV_y = \frac{10m}{3}\sqrt{\frac{gr}{3}} = \frac{10m\sqrt{3gr}}{9},$$

directed downwards.

Checks

- Using projectile motion directly: the particle leaves at height $r + r\cos\theta = \frac{5}{3}r$ with downward vertical speed $v\sin\theta$, where $\sin\theta = \frac{\sqrt{5}}{3}$, so $(v\sin\theta)^2 = \frac{10}{27}gr$. Falling a further $\frac{5}{3}r$ adds $2g \cdot \frac{5}{3}r = \frac{90}{27}gr$, giving $V_y^2 = \frac{100}{27}gr$ again.
- $V_y^2 = \frac{100}{27}gr \approx 3.70gr$ is a bit less than $V^2 = 4gr$, as it must be, since a little of the energy stays in horizontal motion.
- Stepping the motion numerically (with $g = 9.81$, $r = 1.3$, $m = 2$) gives contact lost at 48.19° and vertical momentum 13.745, matching $\frac{10m\sqrt{3gr}}{9}$; the trajectory stays outside the sphere.

Common slips

- Measuring the fall from the floor rather than from the top of the sphere: on the sphere the drop is $r(1 - \cos\theta)$.
- Forgetting that the horizontal velocity is fixed after leaving, and writing $V_y^2 = 4gr$ as if all the kinetic energy were vertical.
- Taking the velocity to make angle $90^\circ - \theta$ with the horizontal; that is the angle of the radius, not the tangent.

Beyond STEP

The condition "contact lost when the reaction vanishes" is a first example of a *constraint force*: a force that adjusts itself to keep a particle on a surface, and which can only push. In Lagrangian mechanics such constraints are built in from the start, and the reaction appears as a Lagrange multiplier. Kleppner and Kolenkow, *An Introduction to Mechanics*, is an excellent place to practise circular motion and energy methods at the next level.

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